FRAUD DETECTION IN ELECTRICITY AND GAS CONSUMPTION

By Dmitrii, Lorenz and Nils

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Objective:

Detect clients involved in fraudulent gas and electricity transactions in Tunisia.

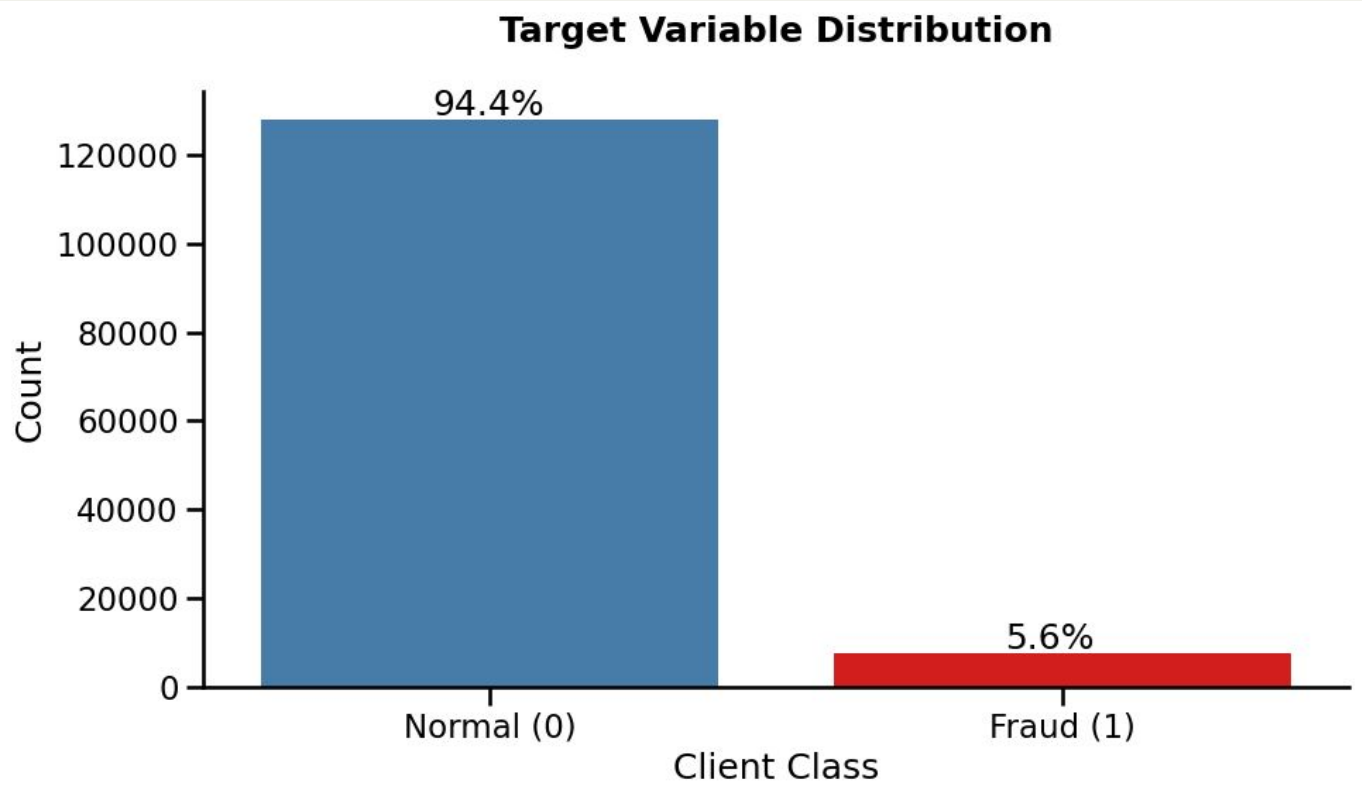
DATASETS OVERVIEW

- **Provided by:** Tunisian Company of Electricity and Gas (STEG) - STEG is a public and a non-administrative company, it is responsible for delivering electricity and gas across Tunisia

- **2 Datasets:**
  - **Invoice dataset:** containing gas- and electricity transactions of all clients
    - **16 features including:** invoice-date, type of tarif, 5 features about the counters, 4 different consommation levels, 2 indizes measuring the consumption, number of months (time span of billed months)
    - **4,476,749 rows**
  - **Client dataset:** containing information about the clients incl. target (fraud (1) or no fraud (0))
    - **5 features:** district and region, client id, client category, creation date
    - **135,493 rows**

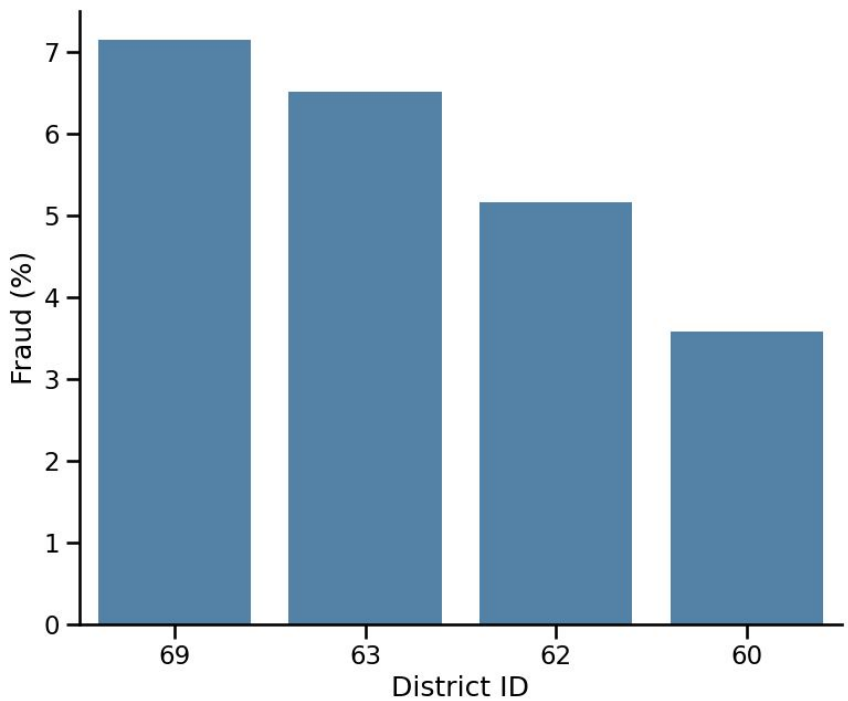
- **Time span of transactions:** 2005 - 2019

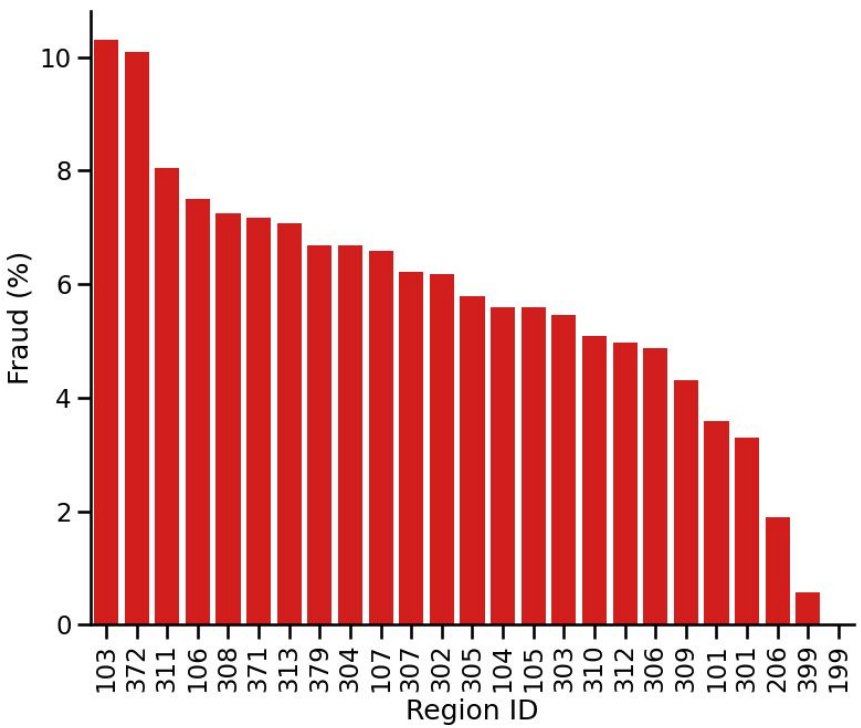
Exploratory Data Analysis (EDA)



Challenge: Highly imbalanced target which could end in a model always focusing on the majority (Normal clients)

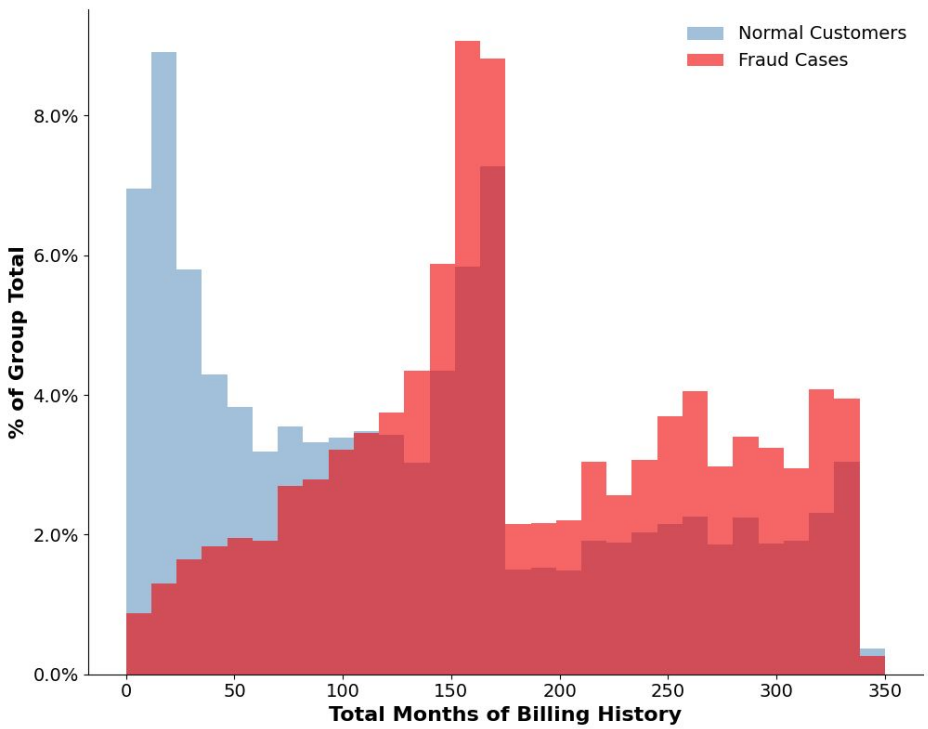
Regions and Districts

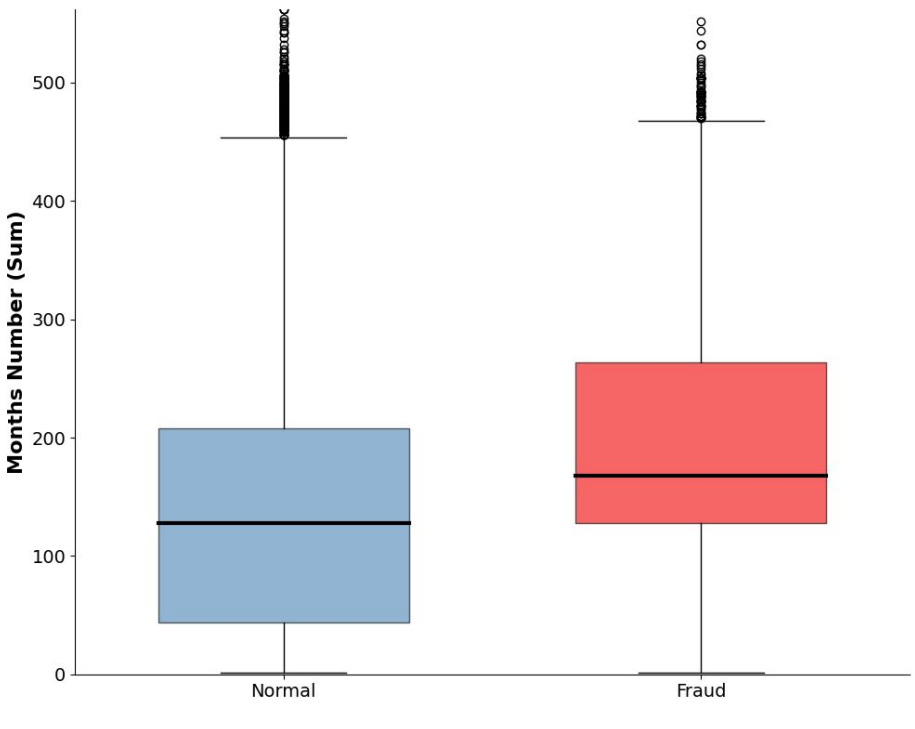


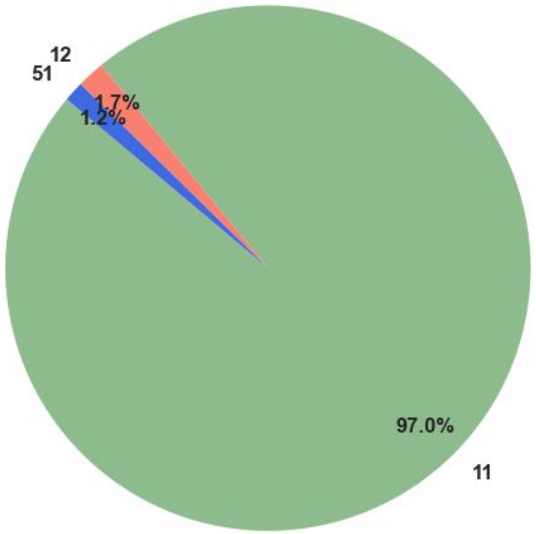


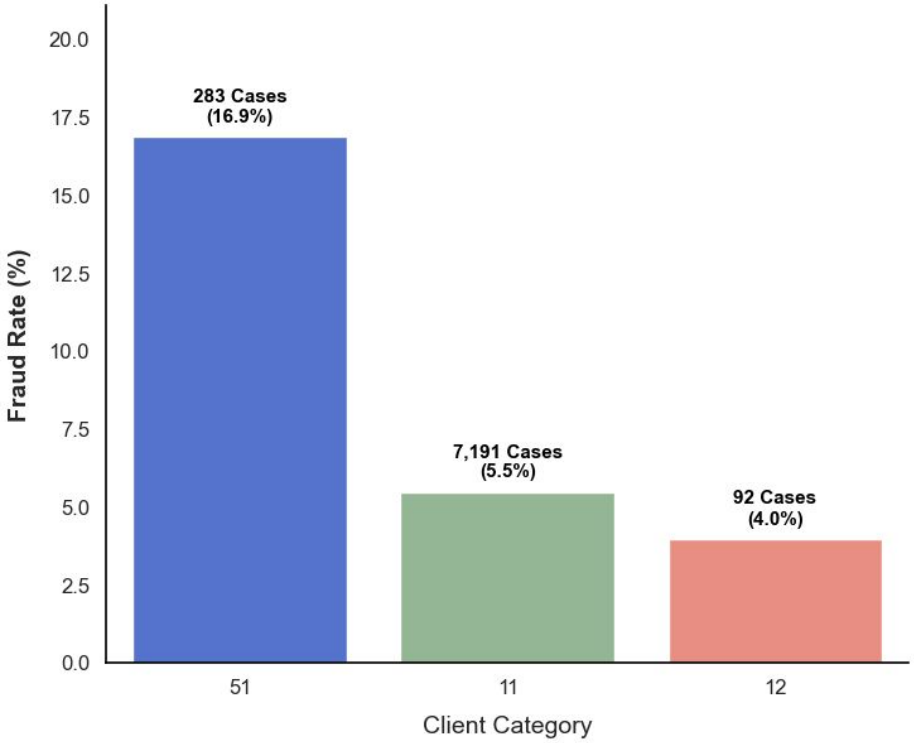
Districts and regions have different percentages of fraud activities

Number of Months Billed



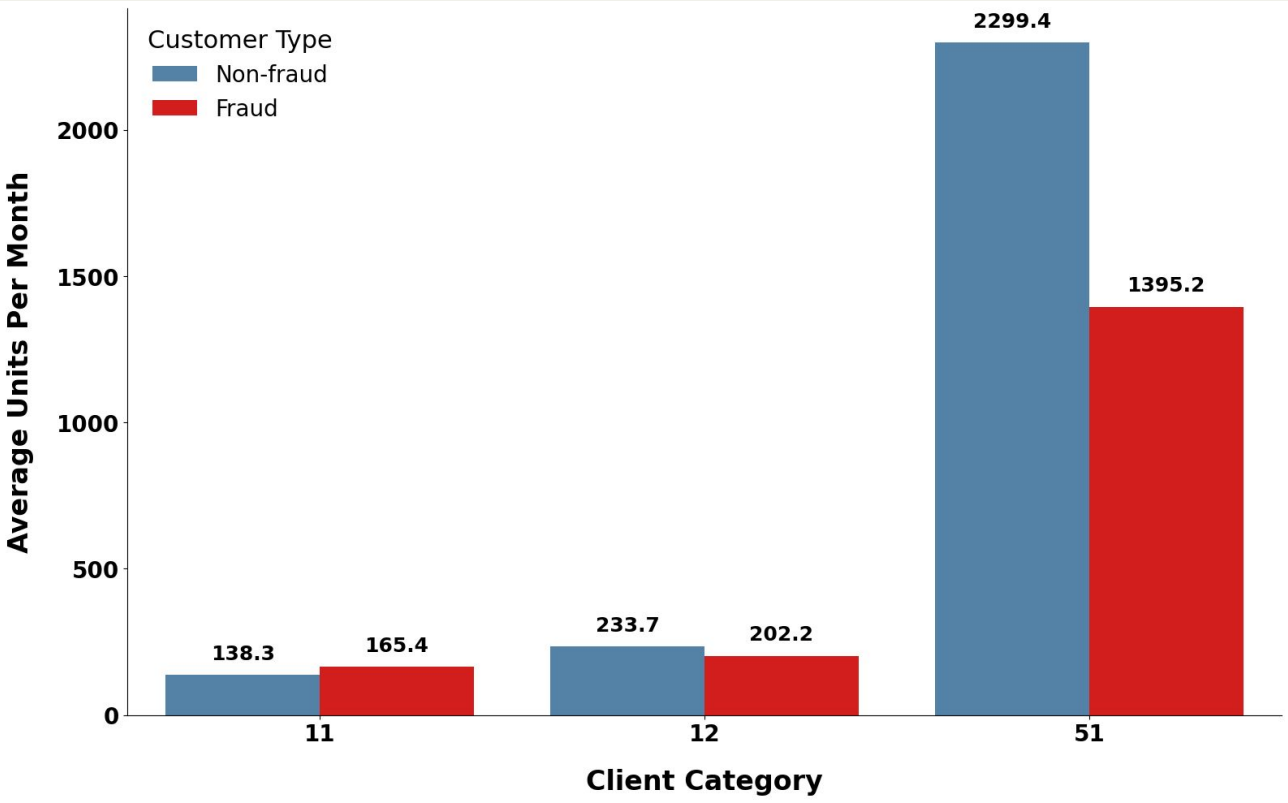






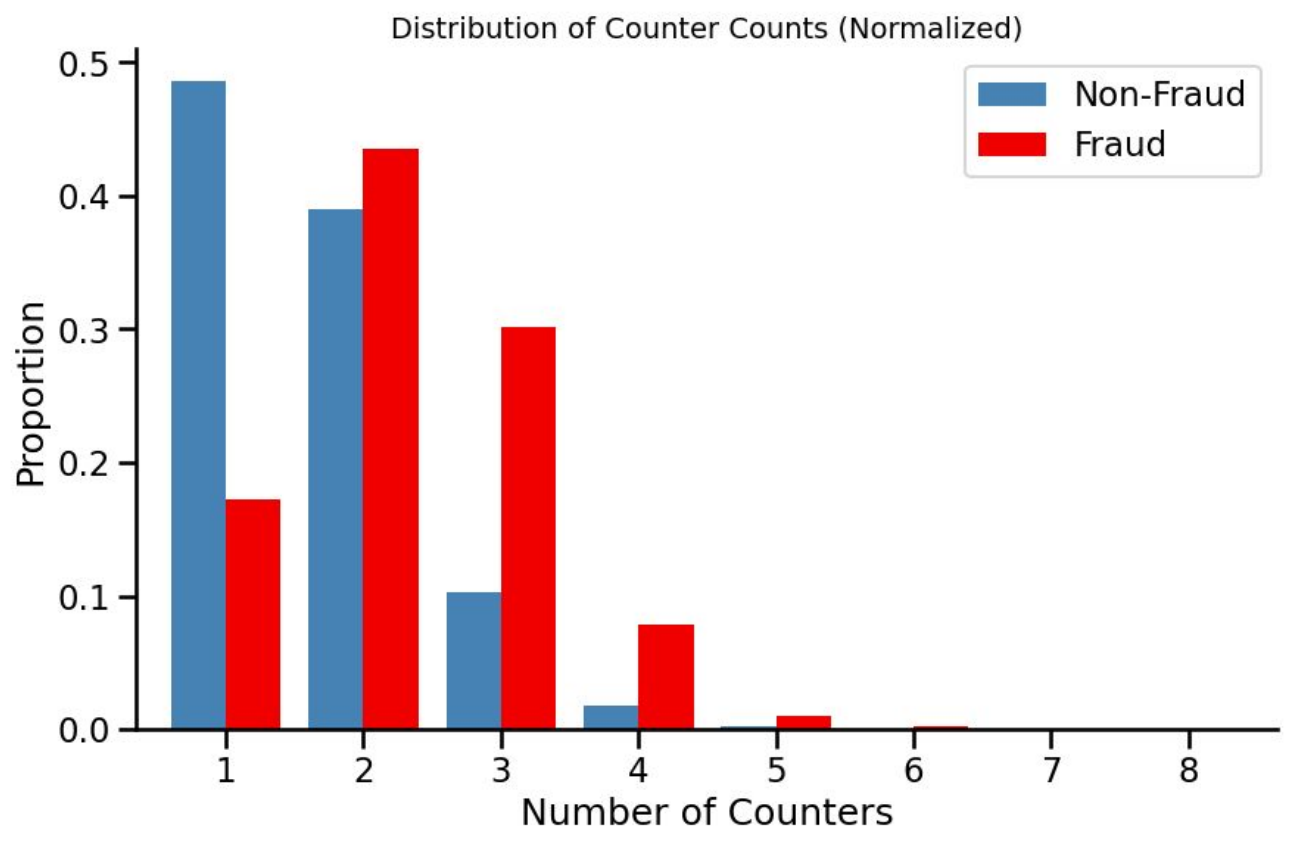
- **Residential (Category 11)**
- **Professional/Commercial (Category 12)**
- **Industrial/Agriculture (Category 51)**

Monthly consumption

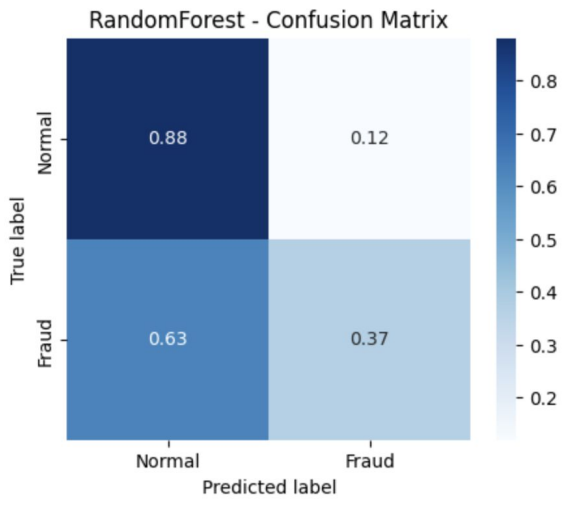


- Residential (Category 11) ➔ low energy consumption
- Professional/Commercial (Category 12) ➔ medium energy consumption
- Industrial/Agriculture (Category 51) ➔ highest energy consumption

Counter Number and Fraud

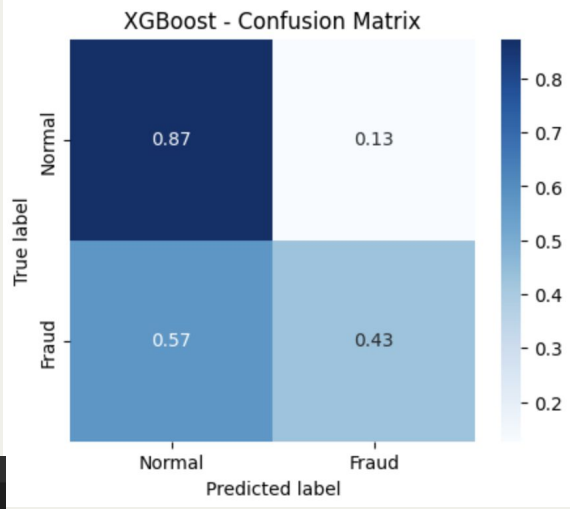


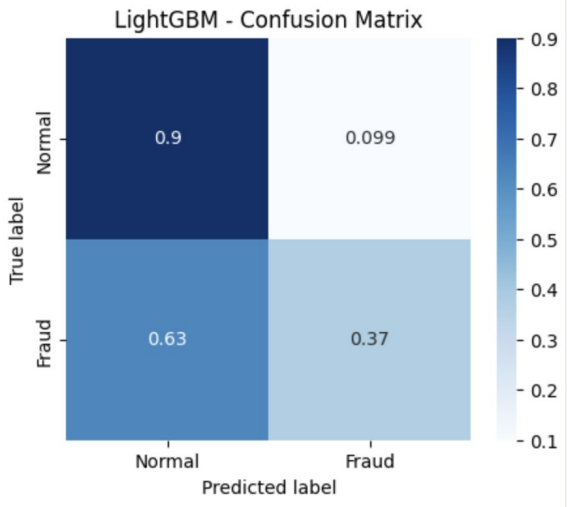
Modelling

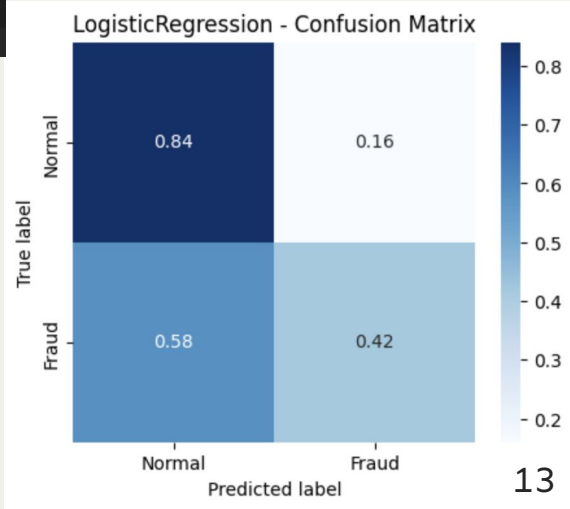


First basic models

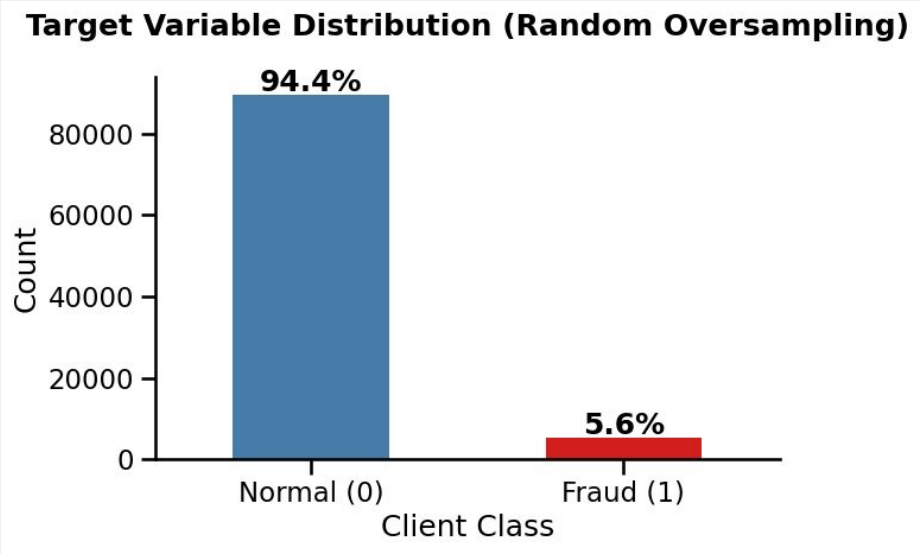
	f1	threshold	precision	recall	roc_auc	pr_auc
LightGBM	0.244810	0.11	0.181935	0.374091	0.758401	0.166704
XGBoost	0.238536	0.10	0.165597	0.426305	0.755808	0.166103
RandomForest	0.216121	0.13	0.153188	0.366821	0.724113	0.137965
LogisticRegression	0.202053	0.08	0.133390	0.416391	0.718909	0.129606

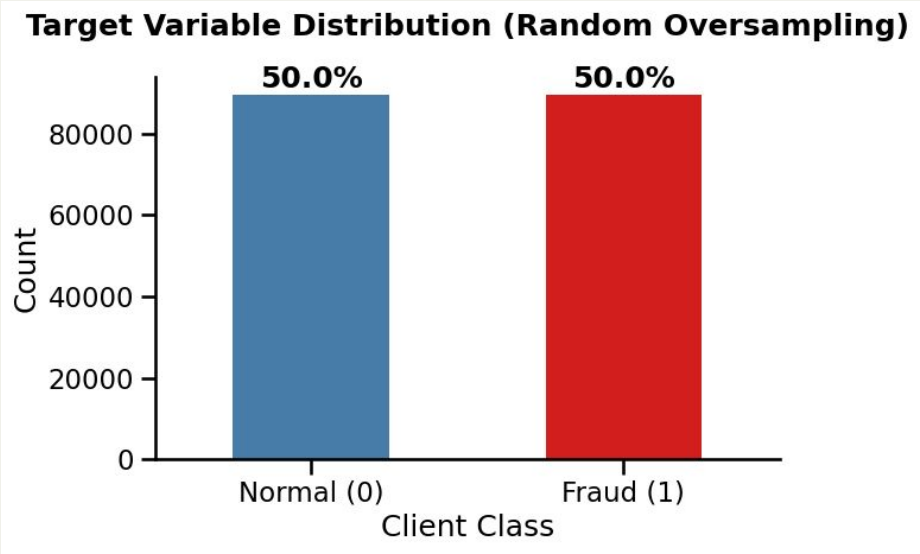


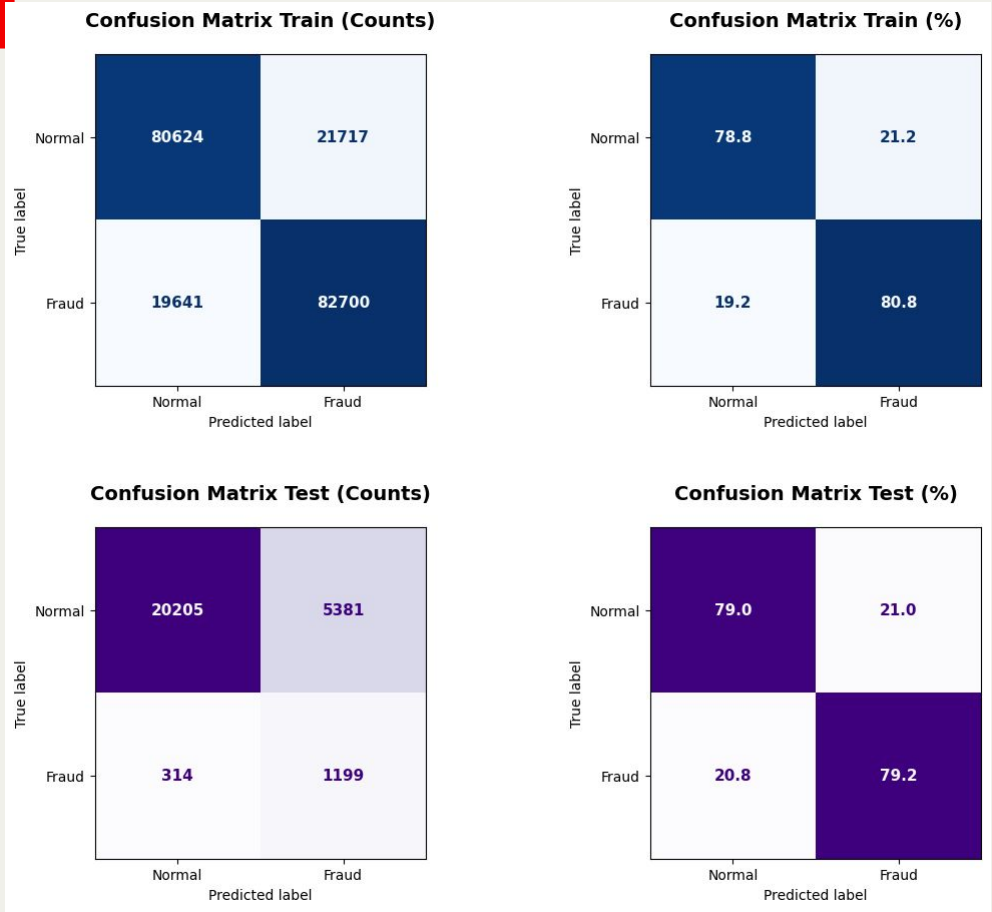




Main challenge: how to handle the highly imbalanced target variable? -> use of Random Oversampling



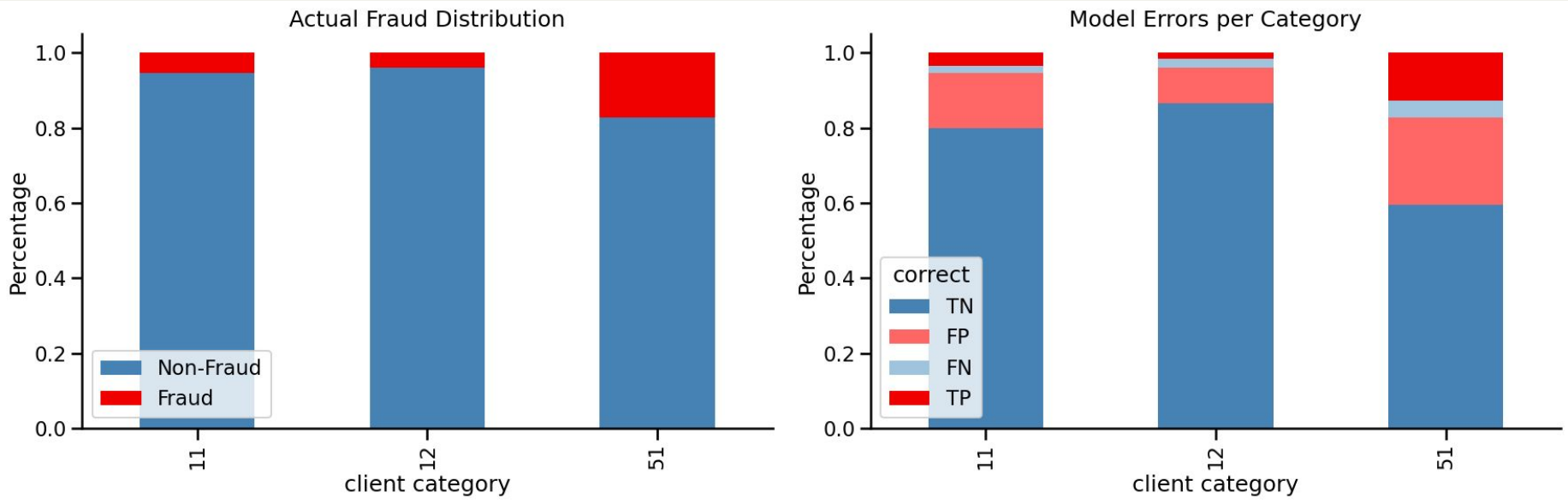




	precision	recall	f1-score	support
0.0	0.98	0.79	0.88	25586
1.0	0.18	0.79	0.30	1513
accuracy			0.79	27099
macro avg	0.58	0.79	0.59	27099
weighted avg	0.94	0.79	0.84	27099

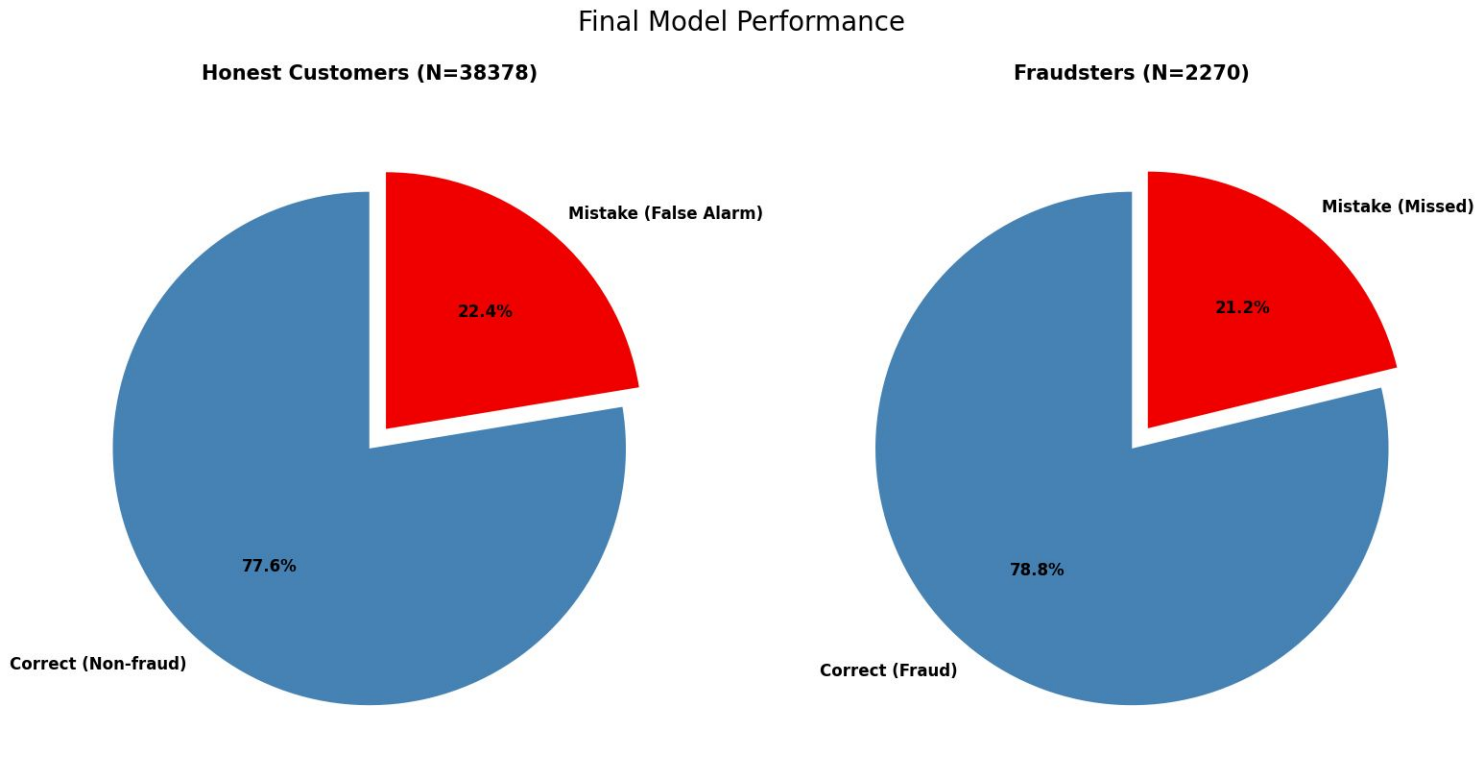
ROC-score: 0.863

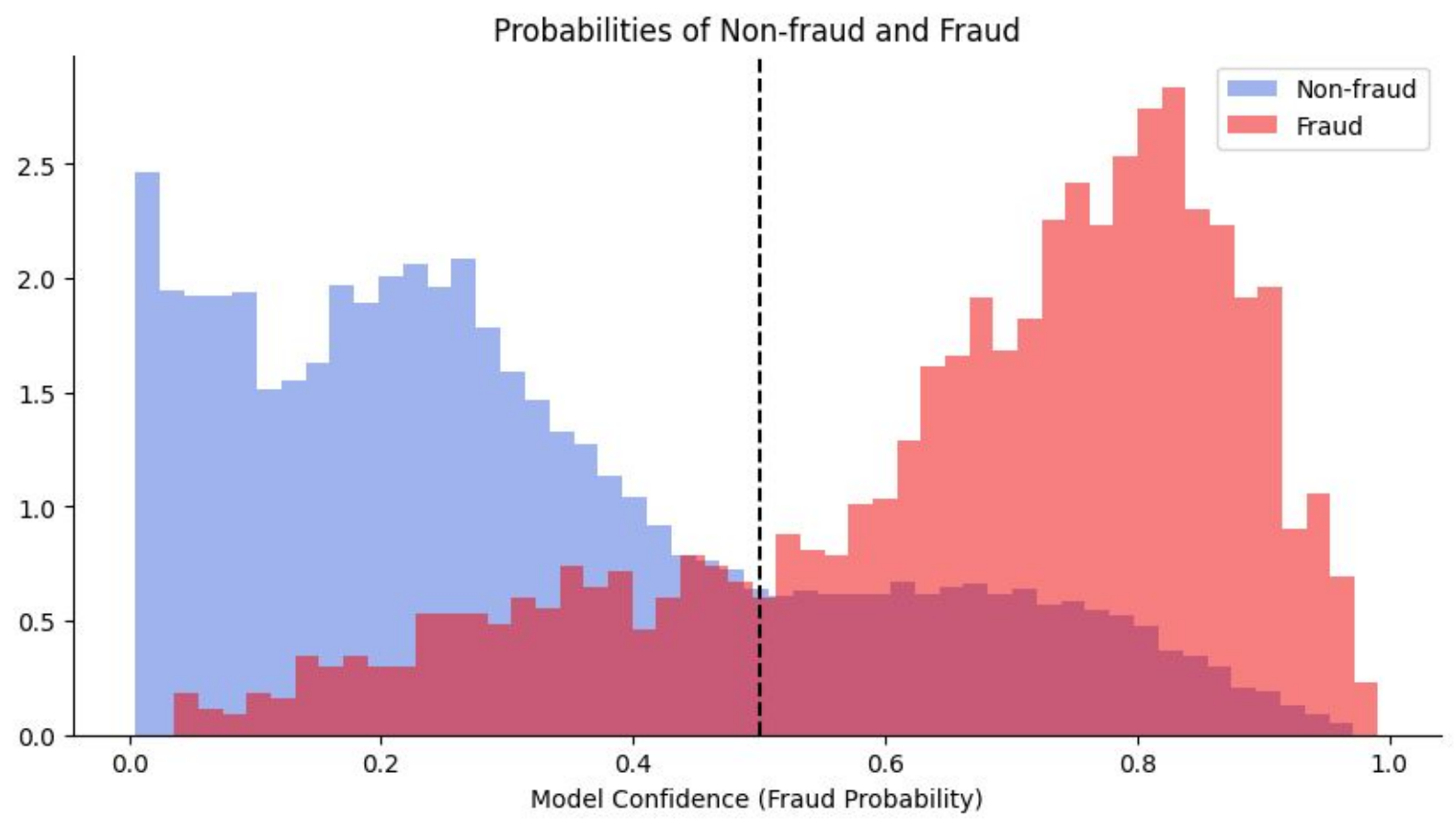
Error Analysis

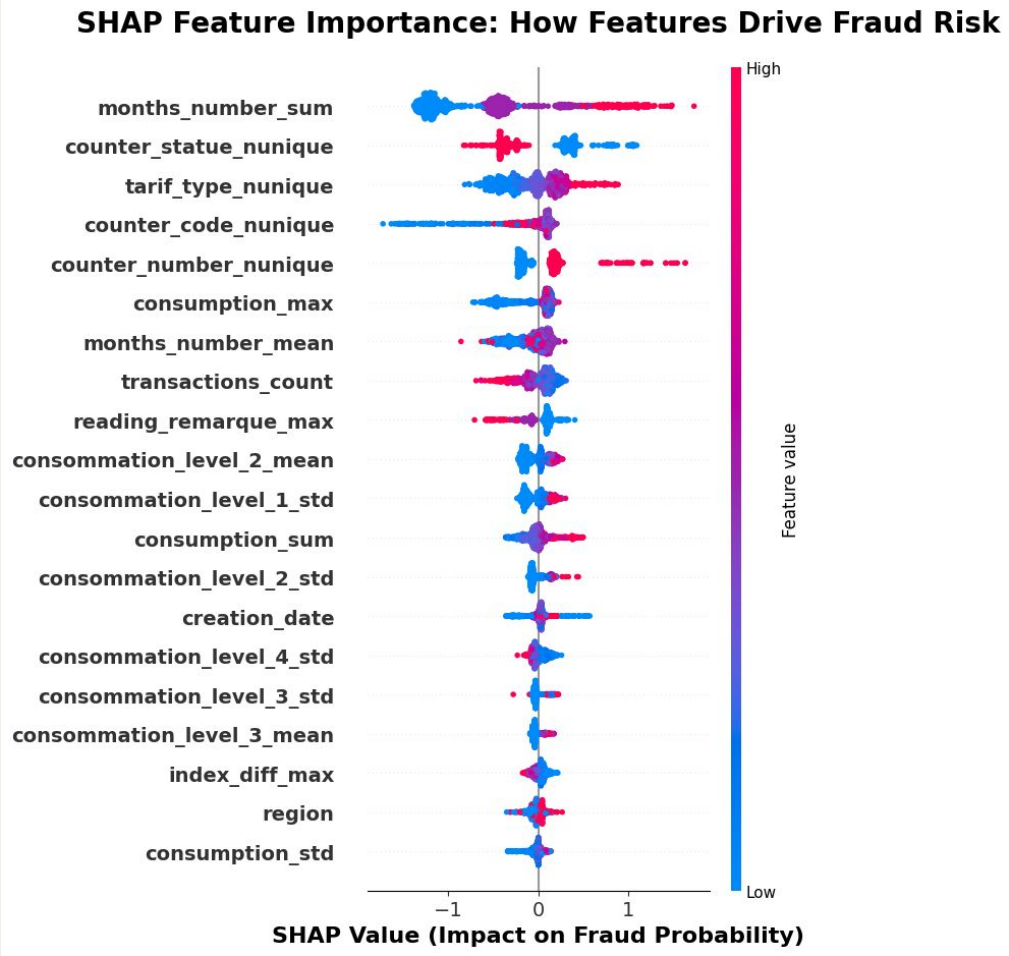

Actual Fraud Distribution
Percentage
1.0
0.8
0.6
0.4
0.2
0.0
Non-Fraud
Fraud
11
12
51
client category
Model Errors per Category
Percentage
1.0
0.8
0.6
0.4
0.2
0.0
correct
TN
FP
FN
TP
11
12
51
client category

THE END

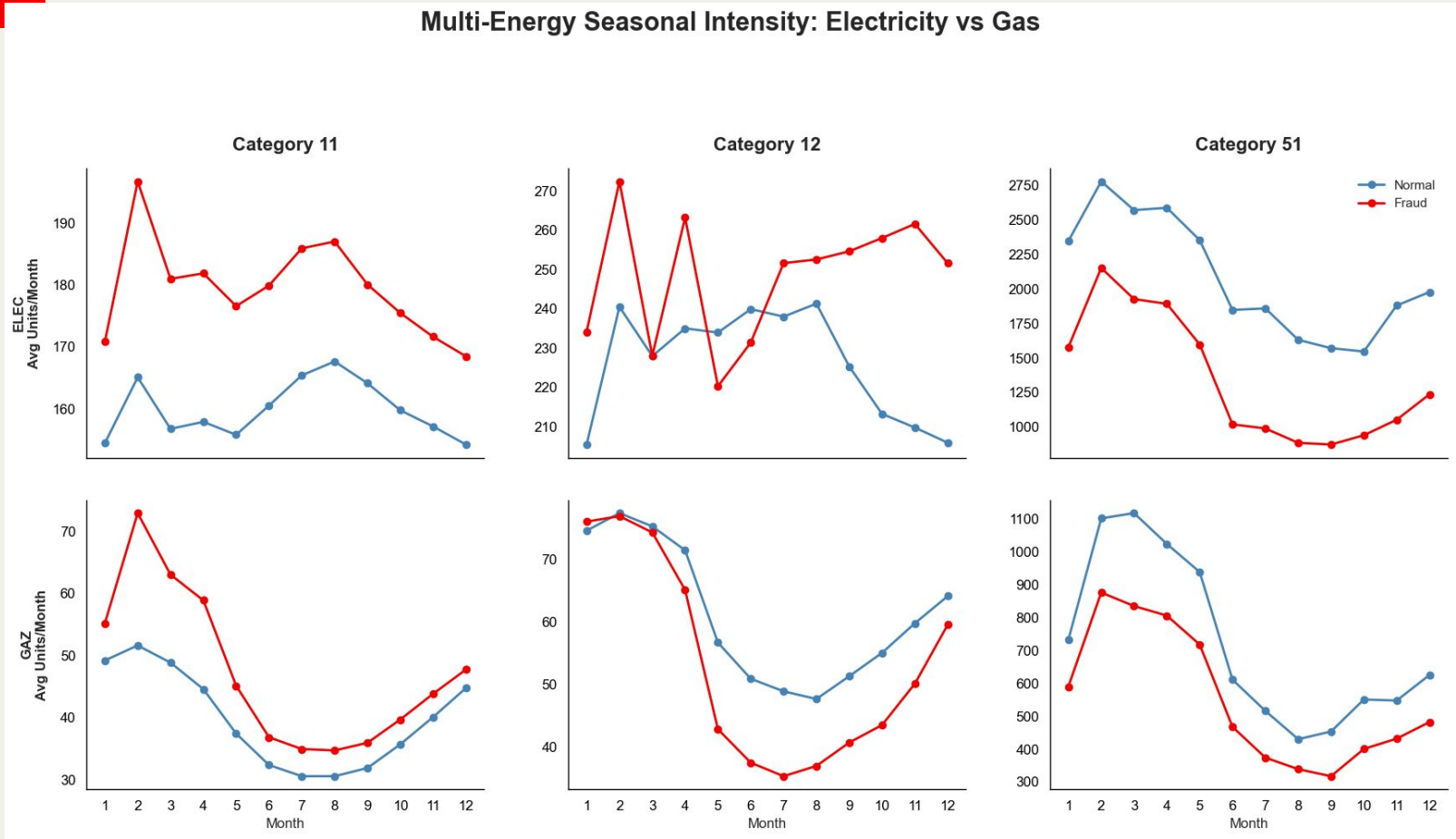
BackUp Slides


Final Model Performance
Honest Customers (N=38378)
Mistake (False Alarm)
22.4%
77.6%
Correct (Non-fraud)
Fraudsters (N=2270)
Mistake (Missed)
21.2%
78.8%
Correct (Fraud)

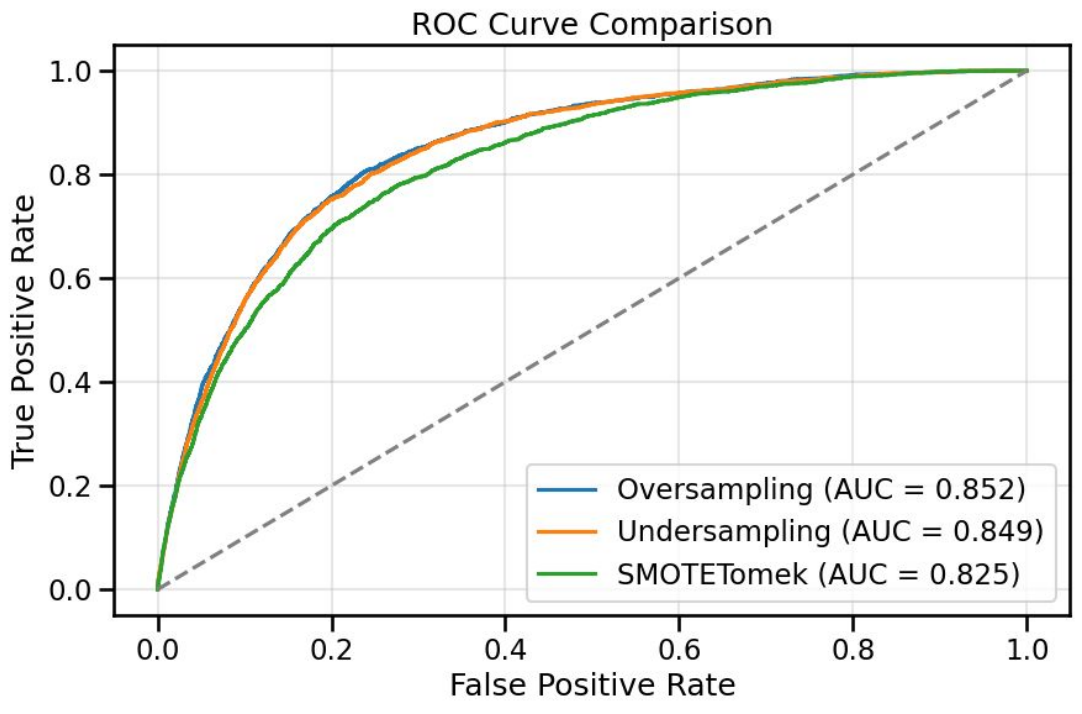

Probabilities of Non-fraud and Fraud
Non-fraud
Fraud
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Model Confidence (Fraud Probability)

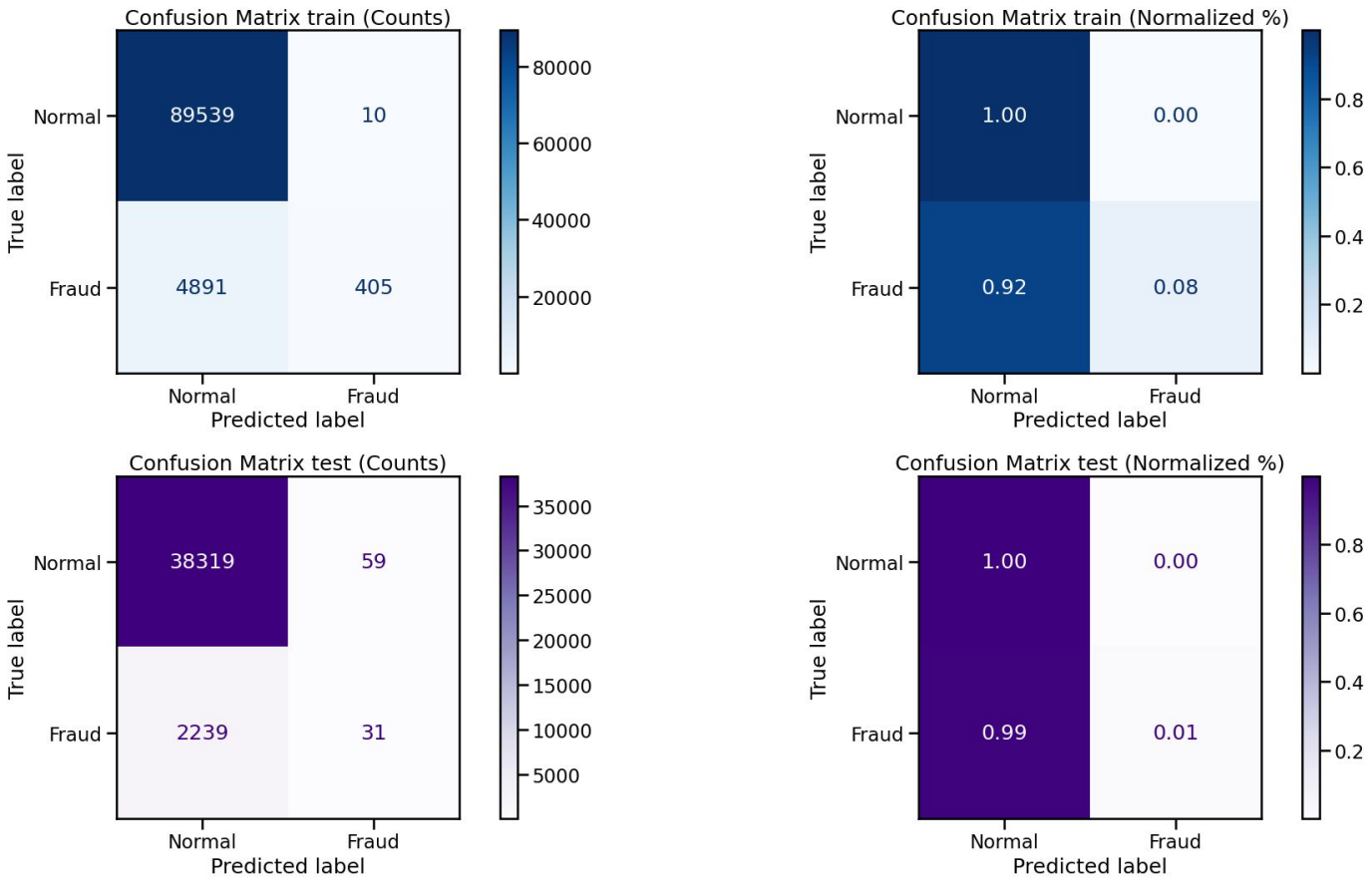

SHAP Feature Importance: How Features Drive Fraud Risk
months_number_sum
counter_statue_nunique
tarif_type_nunique
counter_code_nunique
counter_number_nunique
consumption_max
months_number_mean
transactions_count
reading_remarque_max
consommation_level_2_mean
consommation_level_1_std
consumption_sum
consommation_level_2_std
creation_date
consommation_level_4_std
consommation_level_3_std
consommation_level_3_mean
index_diff_max
region
consumption_std
−1
0
1
SHAP Value (Impact on Fraud Probability)
High
Feature value
Low

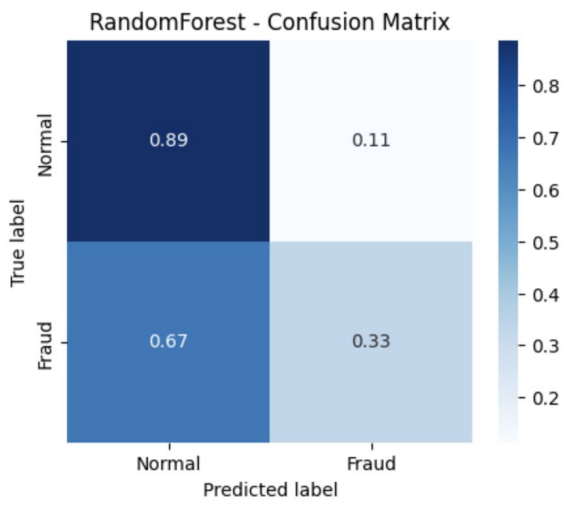
Gas vs. Electricity

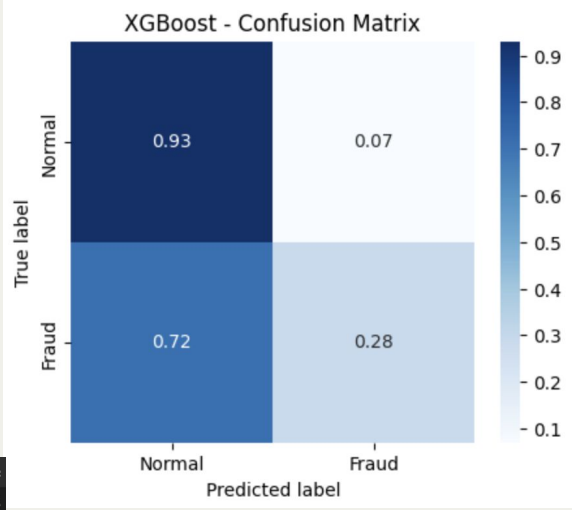
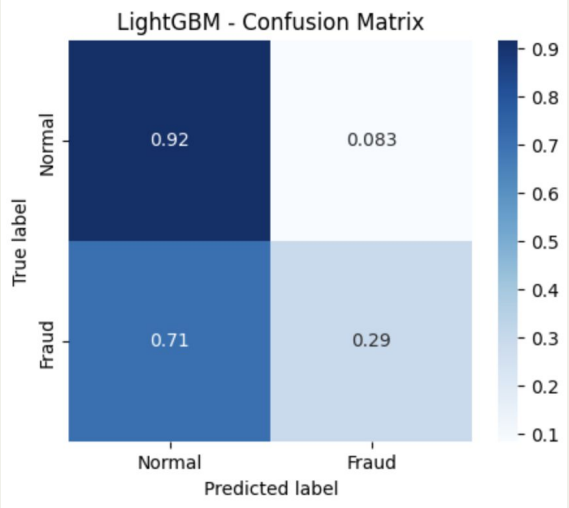
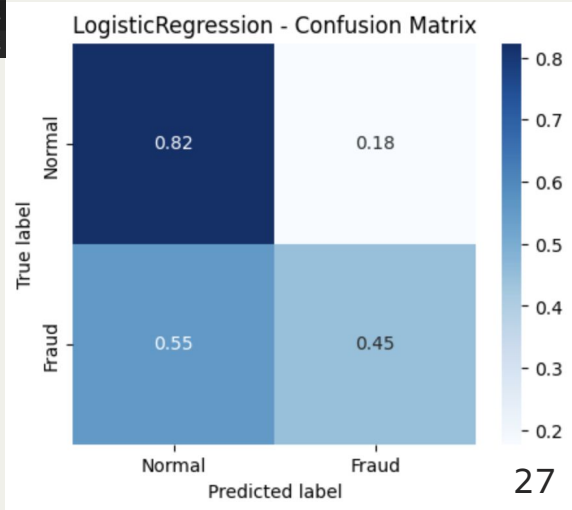


Main challenge: how to handle the highly imbalanced target variable? -> use of different Sampling methods




Confusion Matrix train (Counts)
Normal
Fraud
True label
89539
10
4891
405
Normal
Fraud
Predicted label
80000
60000
40000
20000
Confusion Matrix train (Normalized %)
Normal
Fraud
True label
1.00
0.00
0.92
0.08
Normal
Fraud
Predicted label
0.8
0.6
0.4
0.2
Confusion Matrix test (Counts)
Normal
Fraud
True label
38319
59
2239
31
Normal
Fraud
Predicted label
35000
30000
25000
20000
15000
10000
5000
Confusion Matrix test (Normalized %)
Normal
Fraud
True label
1.00
0.00
0.99
0.01
Normal
Fraud
Predicted label
0.8
0.6
0.4
0.2



With Oversampling

	f1	threshold	precision	recall	roc_auc	pr_auc
XGBoost	0.231409	0.67	0.194846	0.284865	0.743373	0.160078
LightGBM	0.217444	0.61	0.172481	0.294118	0.729359	0.146509
LogisticRegression	0.200923	0.60	0.129658	0.446134	0.724238	0.128068
RandomForest	0.203466	0.18	0.147111	0.329808	0.719667	0.125085